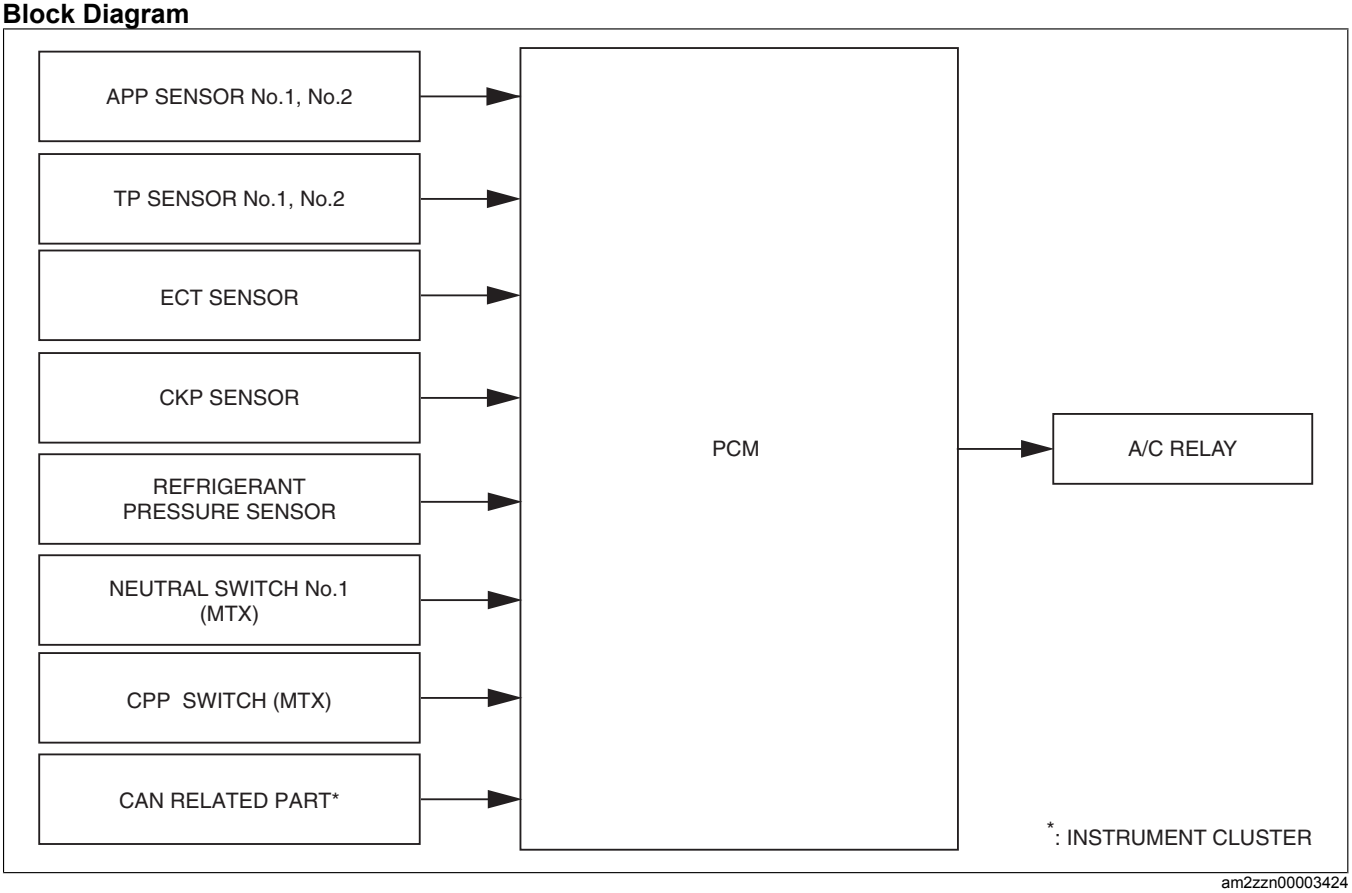
A/C CUT-OFF CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0140q1206200

Outline

- To improve drive performance and A/C compressor reliability, the A/C relay is turned off at the appropriate timing (according to the engine operation conditions), and the A/C operation is controlled.

Block Diagram



am2zzn00003424

Operation

- The PCM stops current to the A/C relay when any of the following conditions are met:

A/C cut-off control operation conditions

—: Not applicable

Operation condition	While A/C relay not energized	Purpose
At engine start	**Approx. 4 s**	Improved startability
When engine is restarted by i-stop control	**Approx. 3 s**	Improved startability
When high engine speed (**approx. 6,700 rpm or more**)	—	A/C compressor reliability assurance
Accelerated	**Approx. 5 s**	Acceleration from standstill/ acceleration performance improvement
Time under i-stop control, vehicle speed of **1 km/h {0.6 mph} or less**	—	i-stop function reliability improvement
Refrigerant pressure **3.14 MPa {32.0 kgf/ cm^2, 455 psi} or more**	—	A/C compressor reliability assurance
Refrigerant pressure of **0.196 MPa {2.00 kgf/ cm^2, 28.4 psi} or less** continues for **5 s or more**	—	A/C compressor reliability assurance
Drive-by-wire control malfunction	—	Reverse driving performance assurance
Panic braking determined	—	Load performance assurance
ECT sensor malfunction	—	Engine protection

Operation condition	While A/C relay not energized	Purpose
When high engine coolant temperature (**approx. 113 °C {235 °F} or more**)	Cuts off the A/C intermittently, and if the engine coolant temperature does not decrease, cuts off the A/C continuously.	Engine protection
Travel on ascending road	—	Ascent performance improvement